far from the production site and then from thereon build a seismically distant DR site. Refer to *Figure 2*.

The major advantage of this approach is that companies don't have to invest in redundant infrastructure and high cost links from day one. As and when business grows and demands another level of protection, DR sites can be built. However, there are some companies especially in the telecom and finance industries, which have invested in a complete DR site from day one. It is the business—rather the business impact in case of disaster—which defines the level of redundancy and justifies the case for two stage DR (multi hop DR sites) or a single stage mirror site.

Also, latencies of the networks involved play a vital role in achieving DR and BC. If the latencies are high and bandwidth is less then replications that are not in sync are generally proposed and implemented. This is also a function of qualification from vendors for different links and the acceptable latencies for their replication technology. While storage based replication software manages the data replication at block level from the storage sub systems, the application servers continue to access data non-disruptively and work on the local copy of data, i.e. the data residing in the storage on the local site. While making a decision for storage based

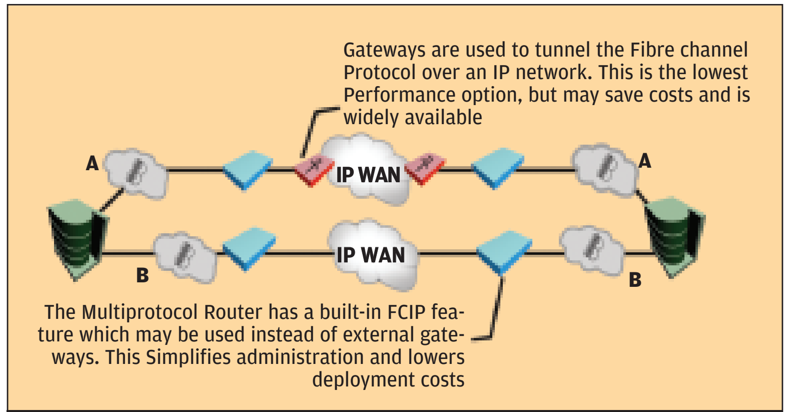

FIGURE 2

utilities and storage sub systems, the organisation should ensure that these replication utilities are certified with the applications or databases they wish to replicate over to another site.

This certification merely ensures that the bits and bytes replicated would be consistent when they arrive at the DR site storage. These utilities are storage controller based and take the I/O load off the servers.

Now, imagine a scenario when disaster strikes. The applications and servers lose connectivity with the production copy of data (there is a high chance that servers are also down). At this moment, the replication utilities start a consistency check on the replicated data (or some of them are already doing it while replicating data) and ensure that each byte transferred is referable by the application servers on the DR site.

Technology	Summary Of Solution And Best-Fit Scenarios
Native Fibre Channel over Dark Fiber	Full performance is required and supported distances is 80 to 100 kilometers or less. Optical cables are in place between the sites, or budget is available to install/lease cabling. Fibre channel switches and/or routers on each end have enhanced buffer-to-buffer credit services to support full-speed operation over long distances, and have Long Wave-Length (LWL) or Extended Long Wave-Length (ELWL) media.
Native Fibre Channel over Passive Coarse wave Division Multiplexing (CWDM)	Same as above, but more inter-site links are required than there are fibers available to carry them. (CWDM supports multiple links per fibre pair.) The LWL or ELWL media must support different frequencies of light in addition to being strong enough to cover the required distances.
Native Fibre Channel over Active CWDM or Dense Wave Division Multiplexing (DWDM)	Like passive CWDM, active WDM services map multiple links to a single fibre pair. They also increase the supported distance by hundreds of kilometers beyond passive technologies, can share links between different protocols such as Ethernet and Fibre Channel, and can be rented from providers. However, they are more expensive than passive solutions.
Fibre Channel Mapped over Synchronous Optical Networks/ Synchronous Digital Hierarchy (SONET/SDH)	It is possible to transport Fibre Channel over SONET/SDH to support very long distances. This is useful when native solutions are unavailable, are cost-prohibitive, or if SONET/SDH networks are already in place.
Fibre Channel over IP (FCIP)	If the other technologies are unavailable, it is possible to transport Fibre Channel over IP. This is a less-reliable and lower-performance option, but it is widely available and typically less expensive.

There is intelligence built into the storage sub systems which allows them to change the state of the data volumes at DR site from read-only to read-write enabled. This is necessary as the copy of data would be the production copy for applications until the primary site comes up.

Geographically dispersed cluster solutions play a vital part in such implementation as they sense the failure on the primary site and then take corrective action by mounting the volumes onto the servers on the DR site and firing up the applications.

The storage subsystem and replication utility ensure consistency; the cluster software along with volume managers ensures that this particular DR copy of data is consistently accessible to the servers and ready to be accessed by applications.

A financial organisation in Mumbai replicates its data to an alternate site, about two kilometers away, and has a full facility where key application users can come and sit on networked PCs and start working. As soon as they log on to the network, their profiles, applications and permissions are restored to that PC and they continue to work seamlessly. Testing has made it possible for this system to work all the time and more so at the time of disaster.

To summarise, many customers in India have got on to DR implementations and few others have successfully completed their projects as well. Implementing DR requires expertise from various areas of IT infrastructure; each one of these functional experts play a vital role in the success of the project.

The key to running a resilient IT setup is to implement DR practices, training the team members and making them aware of their roles and responsibilities. To implement and depend on DR/BC plans, a perpetual test process is absolutely necessary. ■